

## 11.1 Generative Adversarial Nets

Give up on  $p(x)$ . Sample in one pass. Bayes  $D$  is  $1/2$  where  $G$  parked.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

A **GAN** (Goodfellow et al., 2014) is two networks in a contest: a generator that paints fakes, and a discriminator that tries to catch them. You do not write down  $p(x)$  and maximize likelihood. You train a sampler.

**First time this method appears.** A **GAN** is two nets: a **generator** that samples in one pass, and a **discriminator** that scores real vs fake.

**What.** You give up on writing  $p(x)$ .  $G(z)$  maps noise to a fake.  $D(x) \in (0, 1)$  says "real." **Why.**

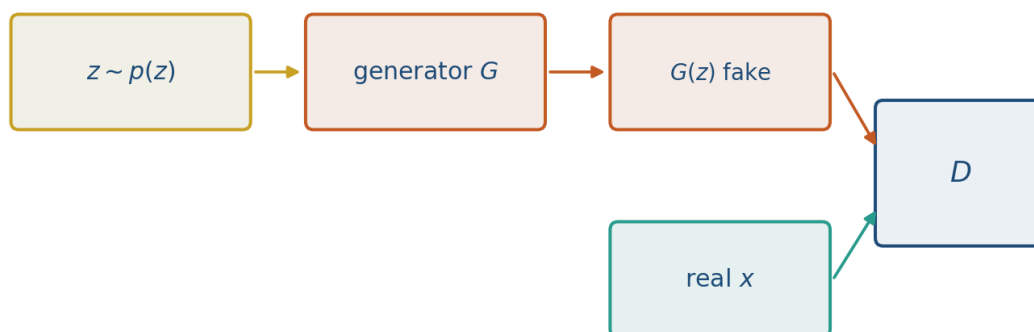
Likelihood models can be blurry or slow to sample. GANs aim for sharp one-pass samples (images, later 11.3's coverage bugs). **Architecture.**  $z \sim p(z) \rightarrow G \rightarrow \text{fake}$ .  $D$  sees reals from the data and fakes from  $G$ . No encoder required. **How.** Train both (details next note). Bayes-optimal  $D$  is  $1/2$  where  $G$  matches the data. **Formula.**  $D^*(x) = p_{\text{data}}(x) / (p_{\text{data}}(x) + p_G(x))$ . At a perfect  $G$ ,  $D^* = 1/2$ .

**Tradeoffs.** + Fast sampling, sharp samples when it works. – No likelihood, unstable train (11.2), mode collapse (11.3), not an LLM method.

### 1. The generator

Noise  $z$  (usually isotropic Gaussian) in; a sample  $x = G(z; \theta_g)$  out, same shape as a real observation.  $G$  is a decoder without an encoder: it does not reconstruct a particular training point. It maps latent draws into data space.

GAN:  $G$  tries to fool  $D$ ;  $D$  tries not to be fooled



The job of  $G$  is to make the **distribution** of fakes look like the distribution of reals. One pretty sample is not a trained generator. If every  $z$  maps to the same pretty point,  $G$  has failed, even if that point fools a human.

In this course  $G$  is a tiny MLP (Lab 11: 8-D noise  $\rightarrow$  2-D point). The same picture is a CNN or a transformer decoder when  $x$  is an image or a spectrogram. The algebra does not change: sample  $z$ , push it through  $G$ , hand the fake to  $D$ .

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## 2. The discriminator

$D(x; \theta_d)$  is a binary classifier: the probability that  $x$  is real. On a real batch it should sit near 1; on  $G(z)$  it should sit near 0. It never sees  $z$ . It only sees the sample.

$$y = D(x; \theta_d) \in (0, 1).$$

Then  $1 - D(x)$  is the probability it calls the point fake.

$D$  is ordinary supervised learning on a moving dataset: reals with label 1, fakes with label 0. The catch is that the fake distribution changes every time you update  $G$ . You never have a fixed classification problem.

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## 3. Why two players

If you trained only  $G$  against a frozen  $D$ ,  $G$  would overfit whatever  $D$  currently likes. If you trained only  $D$ , it would become a perfect detector of yesterday's fakes and give  $G$  a useless (flat) gradient. The pair has to stay in tension. Note **11.2** writes that as a min-max game; note **11.3** is what happens when  $G$  finds one easy fake and stops covering the rest.

For this course, GANs are the first **implicit** generator you meet. Diffusion (Week 12) will give you an explicit noise schedule instead of an adversary. Text-to-image later in the module still needs a way to **condition** (Week 5's CLIP is one).

Draw the figure on the board and leave it up:  $z \rightarrow G \rightarrow$  fake into  $D$ , real  $x$  into  $D$ . Everything later in Week 11 is a comment on that picture.

Stanford CS231N 2025 L13: a GAN is an **implicit** density and a **direct** sample. You give up on  $p(x)$  and keep a one-forward-pass sampler. Diffusion (Week 12) is the other implicit family: iterative, not one shot.

Name what you cannot do:  $p(x)$ ,  $p(z | x)$ , a test log-likelihood. Name what you can do: draw fakes in one forward pass. That pair of sentences is the exam answer for "why a GAN vs why a diffusion model."

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## 4. Teaching this note

About **35 minutes** at the board, then **~8 minutes** of video. This is the first of three GAN notes in a two-hour class; Lab 11 uses the rest of the block.

- **0–12 min.** Implicit vs explicit. You cannot write  $p(x)$  for a GAN. You can sample. Sketch the two-player diagram.

- **12–24 min.** What  $G$  and  $D$  each see.  $G$  never sees a real  $x$  on its own update.  $D$  never sees  $z$ .
- **24–34 min.** Worked 1-D two-mode example. Leave the numbers on the board; note **11.3** will reuse them.
- **Then** play CS231N **46:45–54:00** (GAN setup: give up on density, noise through a generator). Pause when the two-player cartoon appears.

## 5. Worked example

Reals are two spikes in 1-D: half the mass at  $x = -2$ , half at  $x = +2$ . (Lab 11 is the same idea in 2-D, means  $(-2, 0)$  and  $(2, 0)$ .)

Suppose an untrained  $G$  ignores  $z$  and always emits  $G(z) = +2$ .

A Bayes-optimal  $D$  then scores:

| location      | what $D$ sees        | $D(x)$ |
|---------------|----------------------|--------|
| $x = +2$      | half real, half fake | 0.5    |
| $x = -2$      | only reals           | 1      |
| anywhere else | only fakes, if any   | near 0 |

$G$  is “winning” at  $+2$  in the sense that  $D(G(z)) = 0.5$ , a coin flip. It is losing as a **distribution**: the left mode has zero fake mass. One sample at  $+2$  looks perfect. The generator is not trained.

If you now let  $G$  move and it jumps all mass to  $-2$ , the table flips:  $D(-2) = 0.5$ ,  $D(+2) = 1$ . That oscillation is the seed of note **11.3**.



D when  $G$  parks on  $+2$ " src="graphics/11.1-gan-idea/d-bayes.png" />

The optimal discriminator (Goodfellow / CS231N) is  $D^*(x) = p_r(x) / (p_r(x) + p_g(x))$ . Where  $G$  never goes,  $p_g = 0$  and  $D^* = 1$ .  $G$ 's loss never sees that location.

## 6. Where students get stuck

- Treating one pretty fake as success.  $G$  is a distribution, not a Photoshop filter.
- Thinking  $G$  is trained by comparing  $G(z)$  to the nearest real (an autoencoder). It is not. The only teacher is  $D$ 's scalar.
- Forgetting that  $D$ 's data distribution moves. A  $D$  that was perfect last step is scoring a different  $G$  this step.

## 7. Video

Watch [Stanford CS231N 2017 lecture 13, Generative Models](#), 46:45–54:00.

Pause at the two-player setup: noise  $z$  through a generator, no explicit density. Skip PixelCNN/VAE (the first 45 minutes) in class; those are not this note. Notes **11.2** and **11.3** reuse the same URL at later pause points.

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## 8. Practice

1.  $G$  never sees a real image on its own update (only through  $D$ 's scores). How can it learn the data distribution?
2. Why is a perfect  $D$  (always 1 on reals, always 0 on fakes) a problem for  $G$ 's gradient?
3. Name one task where a GAN is a natural fit and one where an explicit likelihood model would be easier to evaluate.
4. In the 1-D example,  $G(z) = +2$  always and reals are equal spikes at  $\pm 2$ . What is the Bayes-optimal  $D(+2)$ ? What is  $D(-2)$ ?
5. You draw 10 fakes and all 10 equal  $+2$ . A classmate says " $D$  is 0.5 there, so  $G$  is done." Write one sentence that disagrees, using coverage.